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GIT 712 (2026) (SAR)

Assignment 1 (submission template)

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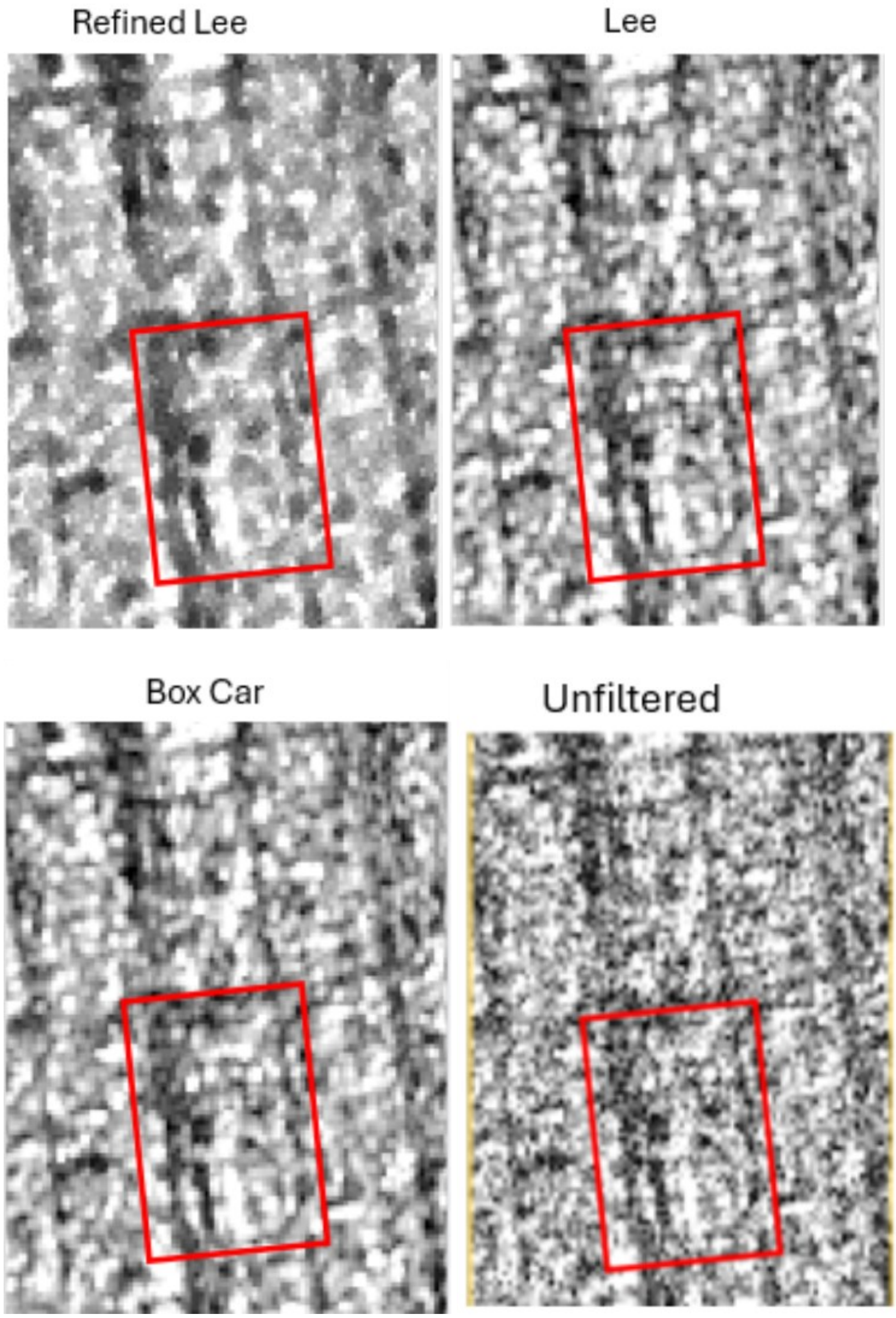
Date:	16/02/2026

Use this template to present and discuss the outputs from the different tasks in the assignment. You will not be directly assessed on the technical finish of this document, but you are expected to present your results neatly and professionally. Include figure and table captions, for example. Make rich use of annotated screenshots to illustrate your work and to support your discussions. The number of words indicated is just a guideline, and is based on the average length from previous years' students. You are welcome to exceed this word length.

Task 1 (TerraSAR-X backscatter processing)

Present, discuss and evaluate the different filtering options you tried. Make use of visual inspection to evaluate the different filters, but you can also include the following five parameters: a) speckle reduction, b) mean value retention, c) preservation of linear features, d) preservation of edges and e) preservation of point features. Include copious screenshots to illustrate your work and support your discussion. Please ensure that the screenshots are sufficiently zoomed in, and of exactly the same extent to ensure that the differences between filters can be clearly seen. [~300 words]

1.1 Speckle Reduction



The area captured in the above images is an urban area which can consist of many widely different backscatter values due to its variable nature.

The unfiltered image shows a large amount of speckle which causes the salt-and-pepper effect, many adjacent pixels or small groupings of pixels have a high variance to the other groups of pixels in the image. The Refined Lee speckle filter has significantly reduced the speckle effect as more neighboring pixels have been grouped and weightily averaged over that region. This creates more sub-regions with clearer edges than the unfiltered image. The image that used the Refined Lee filter has a loss of heterogeneity in the image as the image does appear over-generalized in its averaging of the sub regions.

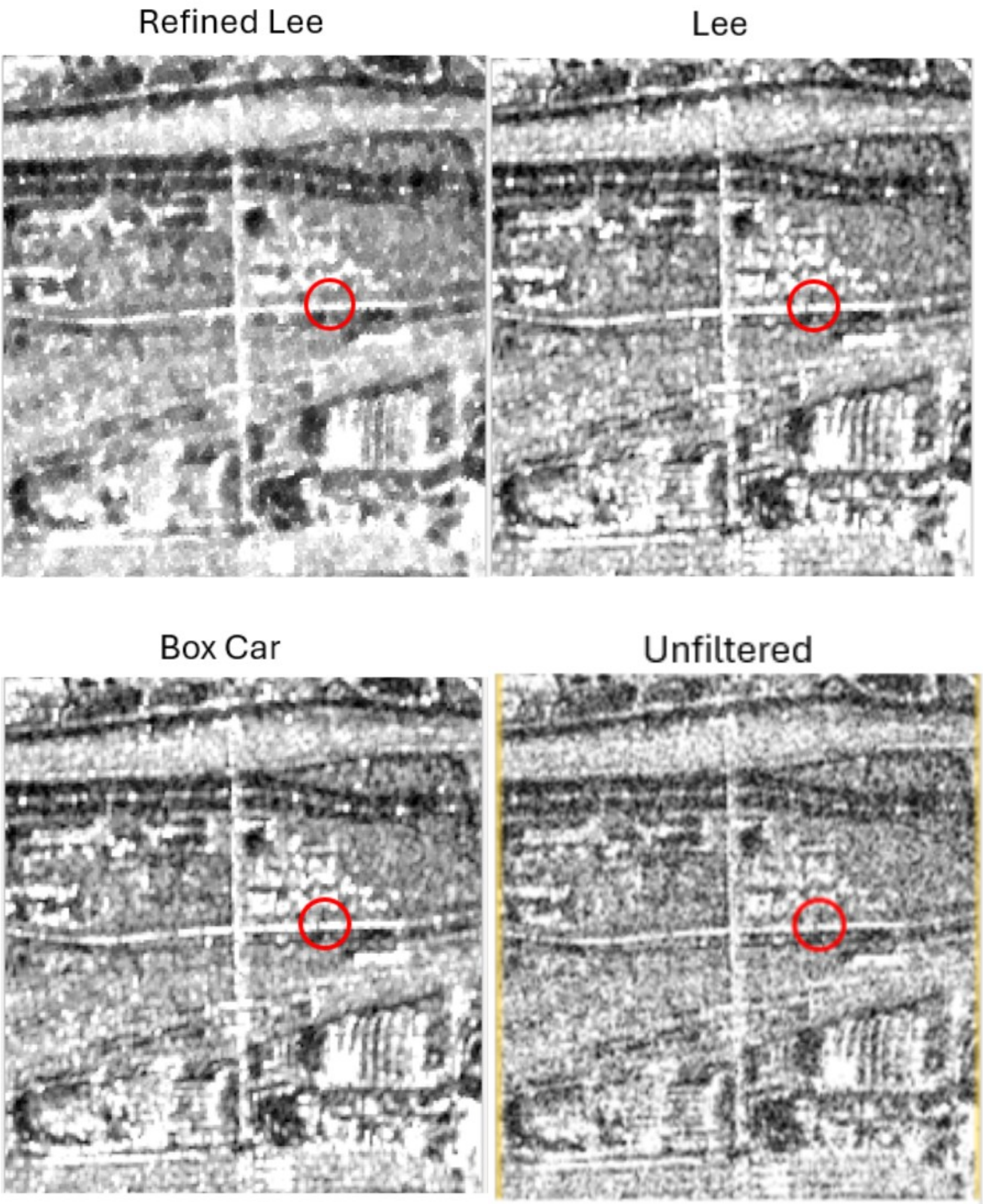
The Lee speckle filter has a severe reduction in the overall speckle of the image when compared to the unfiltered image. There is more speckle in the image than the Refined Lee filter, this shows the Lee filter can retain more heterogeneity in the image while also having less speckle. The Box car Speckle filter is like the Lee filter as it also has reduced the speckle significantly and maintained the same level of heterogeneity as the Lee filter. Areas of a larger backscatter using the Lee and Bocar speckle filtering methods are in larger groups than the Refined Lee filter.

1.2 Mean Value Retention

	Unfiltered	Refined Lee	Boxcar	Lee
Agriculture	-5.805691	-5.044544	-4.957553	-4.959915
Agriculture	-10.35841	-9.481954	-9.373297	-9.424528
Agriculture	-10.543477	-9.66443	-9.564215	-9.614399
Built-up	-0.370395	1.206466	1.76954	1.76954
Built-up	-0.673011	0.975857	1.516877	1.516877
Built-up	-5.556209	-4.310345	-3.789737	-3.789737
Water	-20.654653	-20.037419	-19.872585	-19.9108
Water	-20.913893	-20.297457	-20.14514	-20.174619
Water	-20.731464	-20.037173	-19.857551	-19.89411
Mixed	-11.205312	-10.368697	-10.071885	-10.071885
Mixed	-8.749857	-7.935201	-7.61969	-7.61969
Mixed	-9.979313	-9.081733	-8.741313	-8.741313

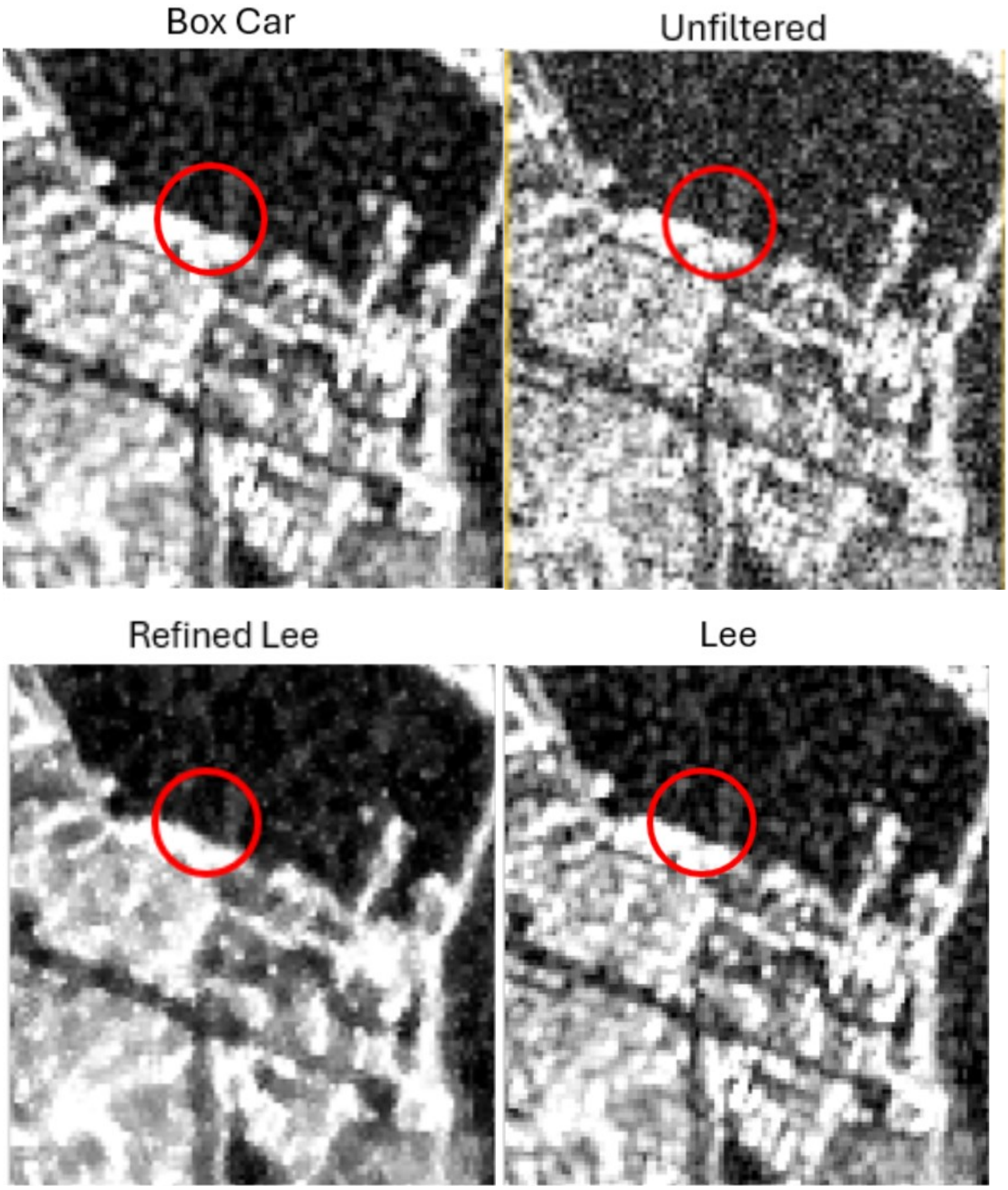
The above table shows values extracted from each filtering method image over various regions in the study area. In every area, the unfiltered image has a lower mean value than the rest of the filtering methods. The Refined Lee filter held the closest mean values to the unfiltered images than Boxcar or Lee filters. This shows that the Refined Lee filter was the best at retaining the mean value over the image. The regions over water bodies maintained their mean value closest to the unfiltered image, this is due to waterbodies backscatter being more homogenous than all other areas. The Lee and Box Car filters were very similar, the Lee filter slightly performing better in retaining the mean values over those regions.

1.3 Preservation of Linear Features



The above images show an intersection of two roads. The unfiltered image maintained a clear distinction of the road from the other land covers surrounding it but lost the detail of the road in the lower and top half of the image. The Refined Lee filter was able to distinguish the linear feature from the surrounding land cover with only some parts of the road segmented. The Lee and Boxcar filters were able to preserve the linear feature equally well but did have gaps in the road due to the speckle left over in the image.

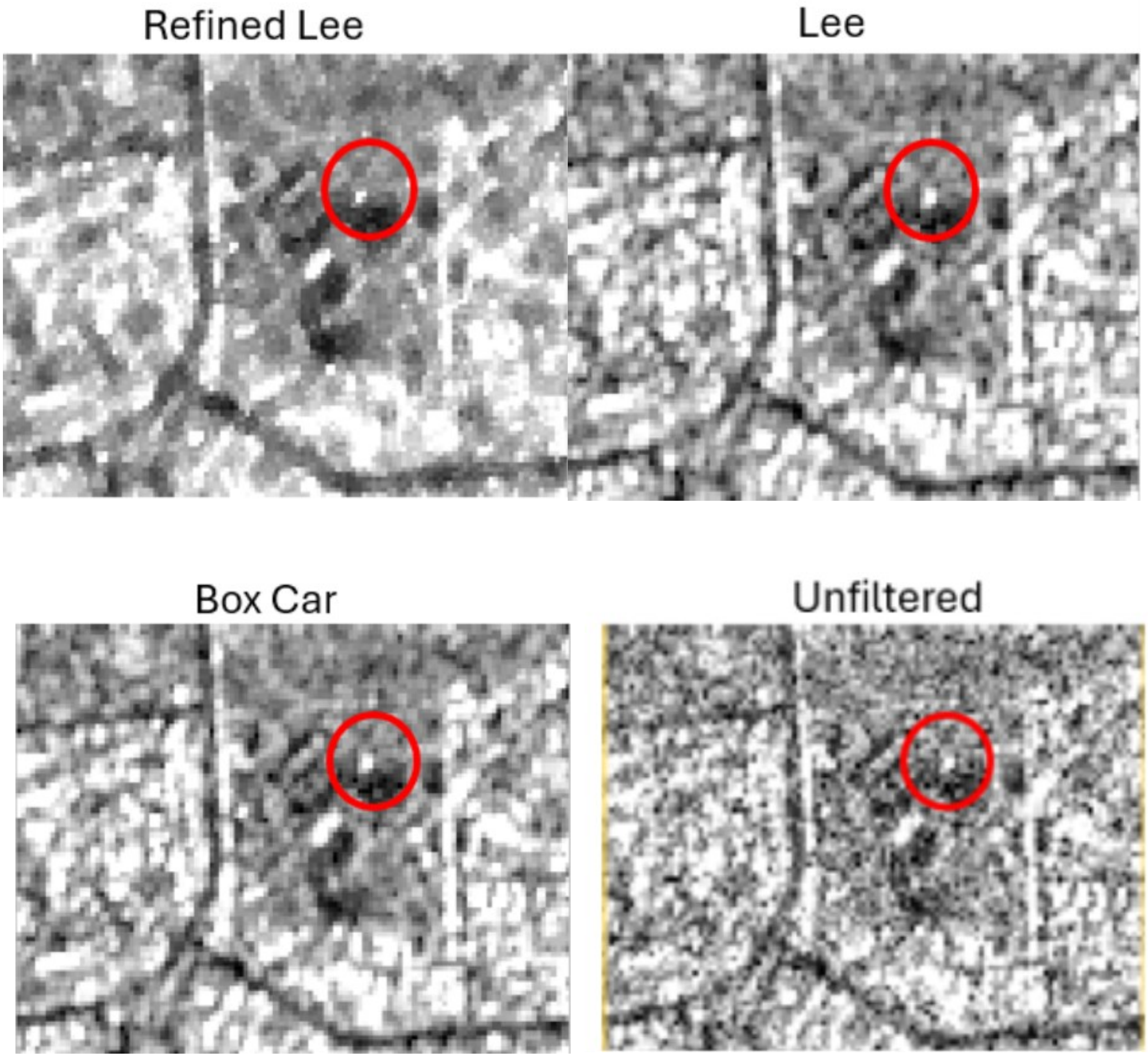
1.4 Preservation of Edges



The above images show the edge of where the water and land start. The unfiltered image provides some distinction between the two features, but the edge is gradually stretched out into the waterbody. This is due to the speckle not being filtered out of the image.

The Refined Lee filter preserves the edge the best out of all the filters, as it shows a clear boundary with minimal smear of the land backscatter into the water. The Boxcar and the Lee filters performed the same, by preserving the edge but also containing more speckle than the Refined Lee which does blur the edge.

1.5 Preservation of Point Features



The above image shows a point feature in the agricultural region. In the unfiltered image, the point is relatively small and is not easily distinguishable from the surrounding areas.

The Refined Lee filter retains the size of the point feature but also makes it easier to distinguish from the surrounding backscatter. The Lee and Boxcar filter performed the same, as they made the point feature more distinguishable from the surrounding backscatter. Where the Lee and Boxcar filters performed poorly is by not preserving the size of the point and rather increasing the size from the unfiltered image.

1.6 Final discussion

Overall, when compared to the Boxcar and Lee filters, the Refined Lee filter demonstrated superior performance in preserving linear, edge, and point features. It also maintained a mean backscatter value closest to the unfiltered data, indicating that it was able to reduce speckle without significant distortion.

Task 2 (RADARSAT-2 multi-temporal filtering)

Present, discuss and evaluate the effectiveness of the multi-temporal filter, relative to the original image and to a single-product filter of your choice. Make use of visual inspection to evaluate the different filters, but you can also include the following five parameters: a) speckle reduction, b) mean value retention, c) preservation of linear features, d) preservation of edges and e) preservation of point features. Include copious screenshots to illustrate your work and support your discussion. Please ensured that the screenshots are sufficiently zoomed in, and of exactly the same extent to ensure that the differences between filters can be clearly seen. [~350 words]

2.1 Speckle Reduction

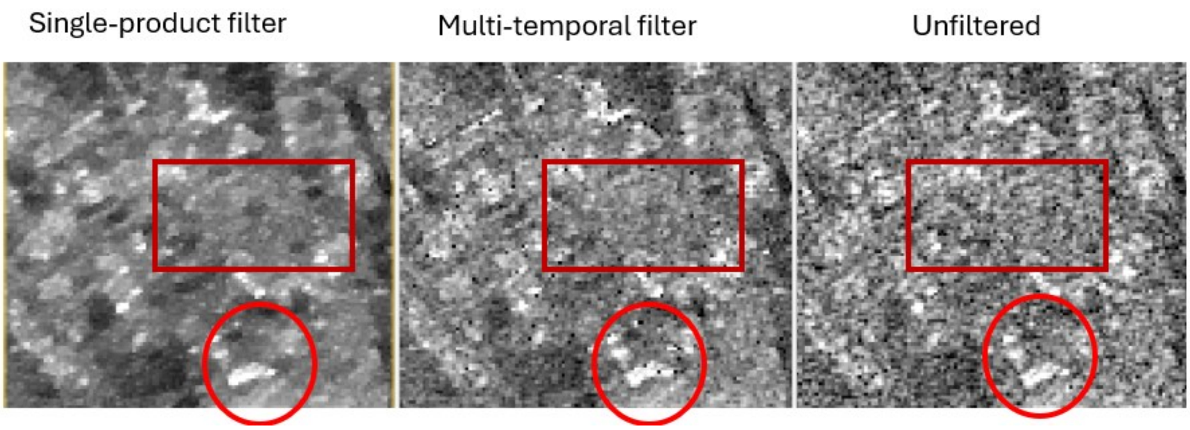
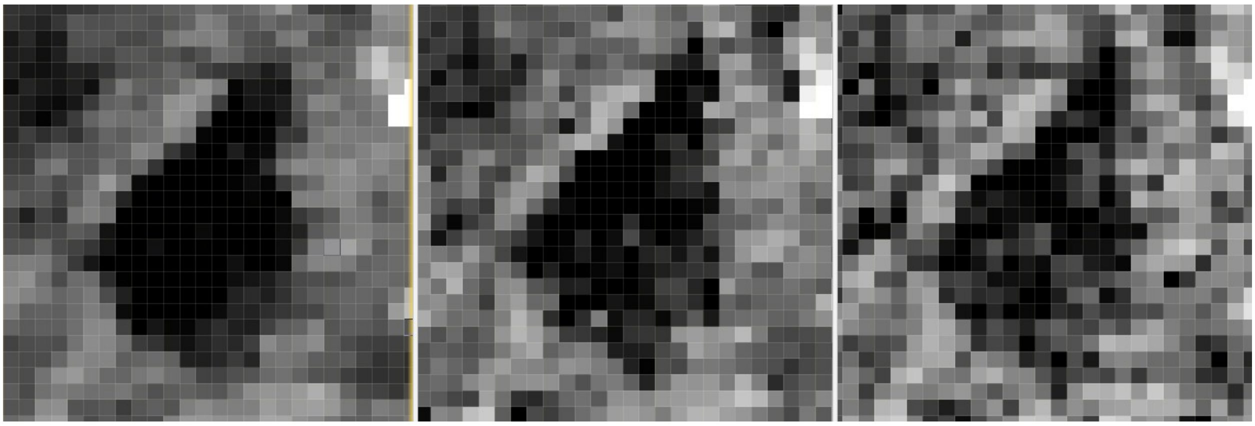


Figure 2.1 Annotated images of a semi-built-up region using different filtering methods.

In Figure 2.1, the Unfiltered image exhibits substantial speckle as seen in the rectangle sample, this makes ground features difficult to distinguish from one another.

The multi-temporal image has a rough landscape texture; the speckle has been significantly reduced from the unfiltered image. This is seen in the sample box as more similar pixels have been averaged allowing for a smoother image without losing significant detail. The Single-product filter reduced the speckle the most, which creates a 'waxier' look to the image when compared to the multi-temporal filter. This does sacrifice detail preservation as more pixels are assigned the same or similar values.



2.2 Mean Value Retention

	Unfiltered	Single	Multi
Agriculture	-24,944071	-24,352	-24,868
Agriculture	-23,819732	-23,125	-23,267
Agriculture	-24,702119	-23,66	-24,257
Built-up	-11,920445	-11,168	-11,705
Built-up	-12,217129	-11,421	-11,985
Built-up	-12,476741	-11,687	-12,205
Water	-11,674068	-10,906	-11,494
Water	-9,505964	-8,4186	-9,0205
Water	-4,614537	-3,6916	-4,6652
Mixed	-5,456413	-4,3268	-5,2928
Mixed	-14,673707	-13,859	-14,404
Mixed	-8,298194	-7,6818	-8,2102

Figure 2.2 Table showing the mean values of sub regions per filter type.

In Figure 2.2, the Multi-temporal filtered image retained the closest mean value to the Unfiltered image than the Single-product filter. This is due to the multi-temporal filters ability to use information from the stack of images rather than use neighbouring pixels to determine the pixels new value.

2.3 Preservation of Linear Features

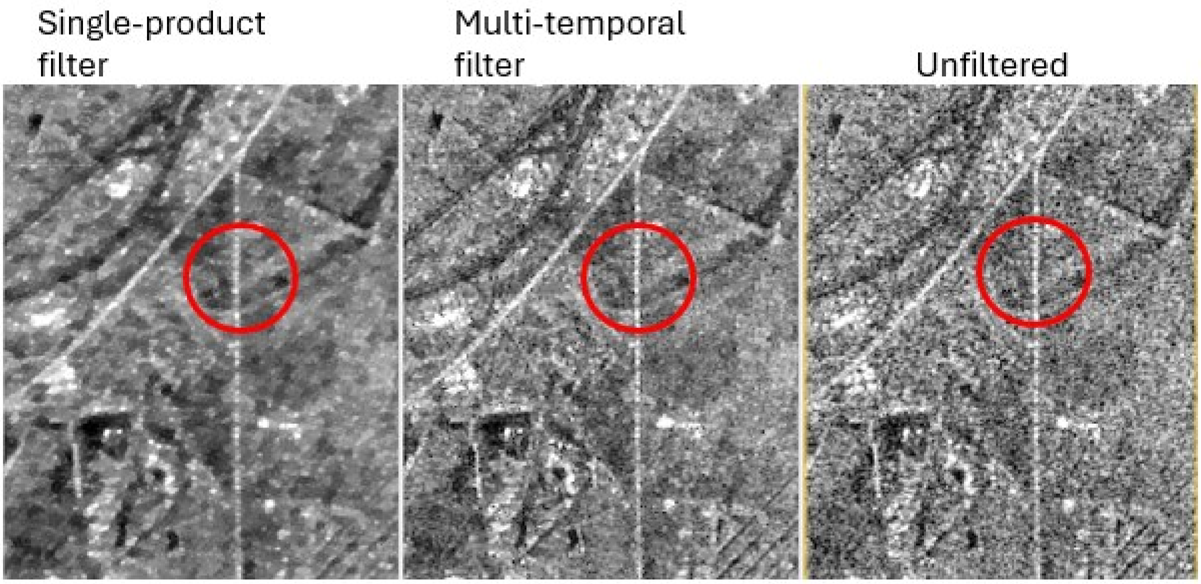


Figure 2.3 Images of a railroad segment.

The Unfiltered image in Figure 2.3 depicts a railroad segment, this linear feature contains substantial speckle which obscures the linear feature to have an irregular shape. The Single-product filter was able to distinguish this rail segment but due to the edge smearing within the image it makes the linear feature discontinuous. This is seen in the highlighted circle as the rail appears to end in certain areas. The multi-temporal filter performs the best at preserving this linear feature, as the edges of this feature don’t smear into the surrounding features while also maintaining the continuous nature of the railroad.

2.4 Preservation of Edges

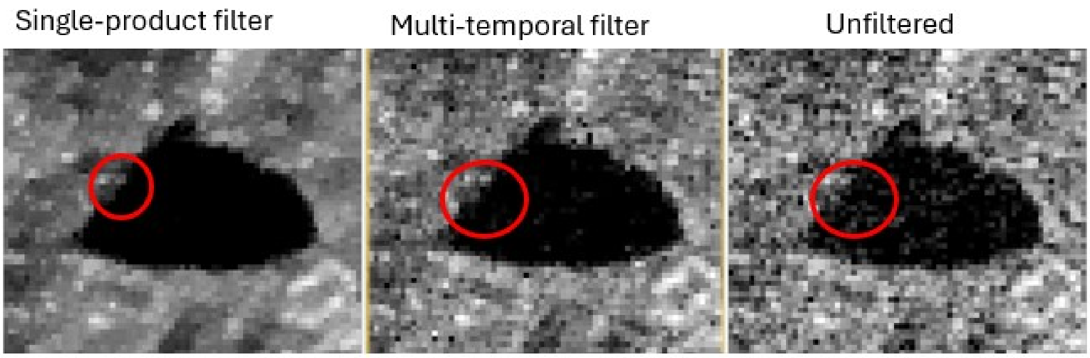


Figure 2.4 Images of a dam across different filtering methods.

The unfiltered image in Figure 2.4 exhibits a substantial amount of edge smearing on the sides of the dam as seen in the highlighted circle. The multi-temporal filter is like the single-product filter, as the edges of the dam have been preserved with very minimal edge smearing. The multi-temporal filter does exhibit slightly more speckle on other edges of the dam, this variability is likely due to the time-series nature of the data, which captured the dam at varying stages of fullness across the different acquisition dates.

2.5 Preservation of Point Features

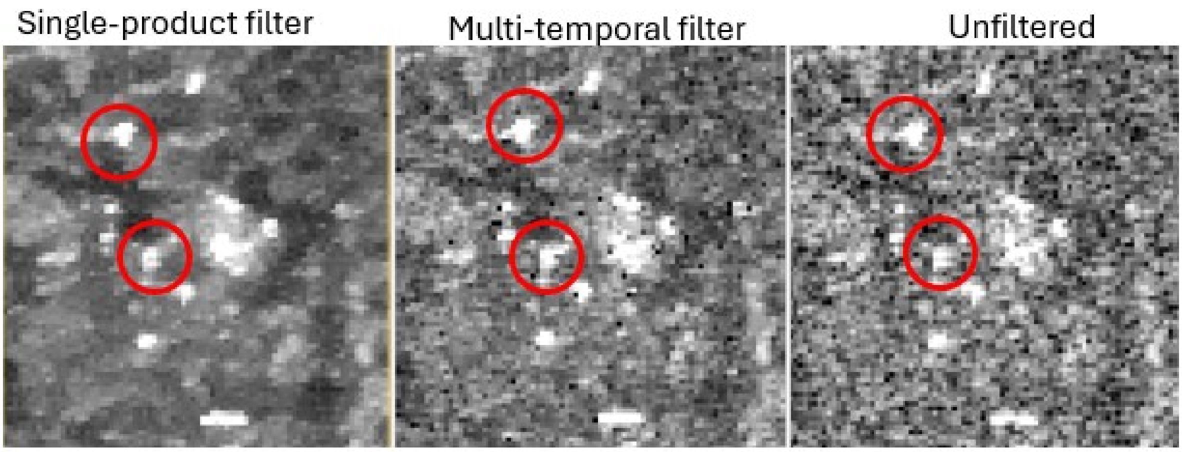


Figure 2.5 Images of point features across different filtering methods.

The unfiltered image in Figure 2.5 shows point features within the highlighted circles, these point features are difficult to distinguish from the surrounding areas as the surrounding backscatter is merging with the point features. The highlighted circle in the single product filter exhibits a reduction in size of the point feature when compared to the multi-temporal and unfiltered images. The single product filter performed better at isolating these point features from the surrounding areas compared to the multi-temporal filter but does lose detail of the point feature. This is where the multi-temporal filter outperforms the single product filter in its ability to isolate the point feature while not losing detail.

2.6 Final discussion

Following a comparison between single-product and multi-temporal (time-series) filtering, the results indicate that the single-product approach is more effective at speckle reduction and edge enhancement, this comes at the cost of some spatial detail. The multi-temporal filter successfully reduced noise while maximizing the retention of fine-scale information and feature retention. This shows that the selection of a speckle filter is application dependent. Single-product filters are required when radiometric smoothness and sharp boundaries are required without the temporal aspect of the data. However, multi-temporal filters perform better when the objective is to preserve temporal trends and spatial detail while achieving a moderate reduction in speckle.

Task 3 (Sentinel-1 full scene dual-pol backscatter)

Discuss the backscatter patterns you see in the scene. Consider sensor and ground parameters, and comment on water, urban areas, agriculture, steep terrain, roads, railways, bare areas, backscatter variations, unexpected or unexplained phenomena, interesting features etc. Consider experimenting with different RGB combinations of the VV and VH bands. [~400 words]

3.1 Agriculture

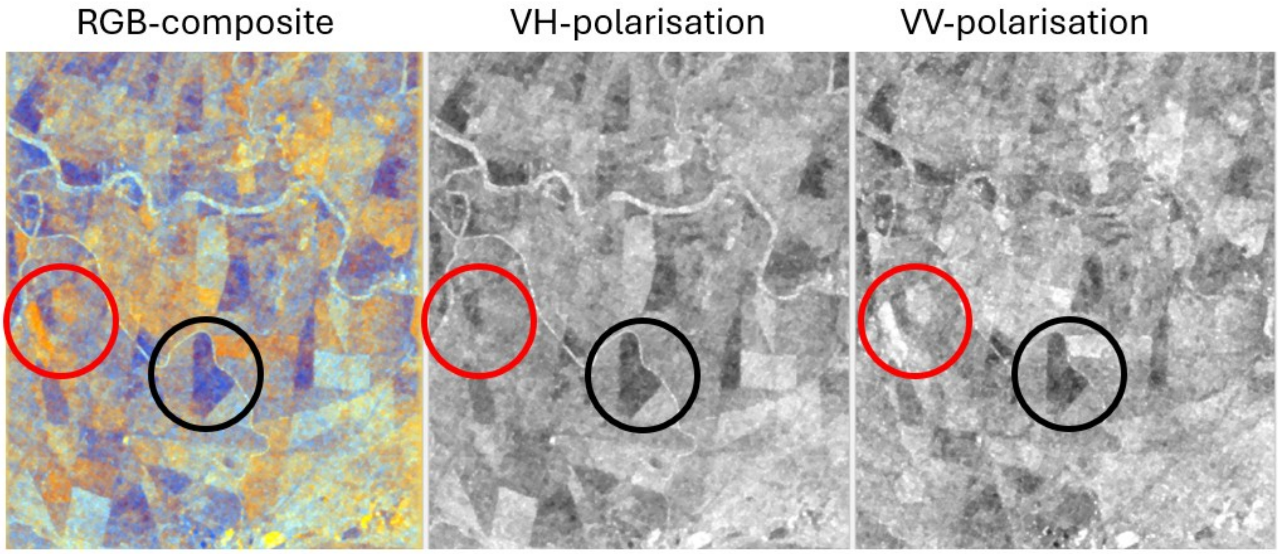


Figure 3.1 The different polarizations of agricultural land cover.

Within the scene, the agricultural areas exhibit an array of different backscatter intensities. As the C-band of the Sentinel 1 sensor is sensitive to rough ground surfaces, this means that high-backscatter areas are likely to be recently ploughed or smoother regions will have less scattering. The area highlighted by the red circle in the VV-polarization shows a bright patch of land, alternatively, for that same region in the VH-Polarization the area doesn't display the same intensity of brightness. This is interesting as VH-polarization exhibits large amounts of volume scattering, but the VV-polarization contains high-backscatter. This means that these areas with high-backscatters in the VV-polarization are likely to be soils with high soil moisture content or rougher surfaces. The C-band has a 5.6cm wavelength and it generally interacts with the canopies of these agricultural areas. The area highlighted by the black circle exhibits minimal backscatter in both polarizations which possibly means that the surface is very smooth with minimal vegetation which will exhibit more specular reflection. The false colour composite was used in the RGB image, the red circle in the RGB image further emphasizes that it is soil moisture and surface roughness rather than vegetation as it is green and all vegetation is mapped in the blue or red channels.

3.2 Roads and railways

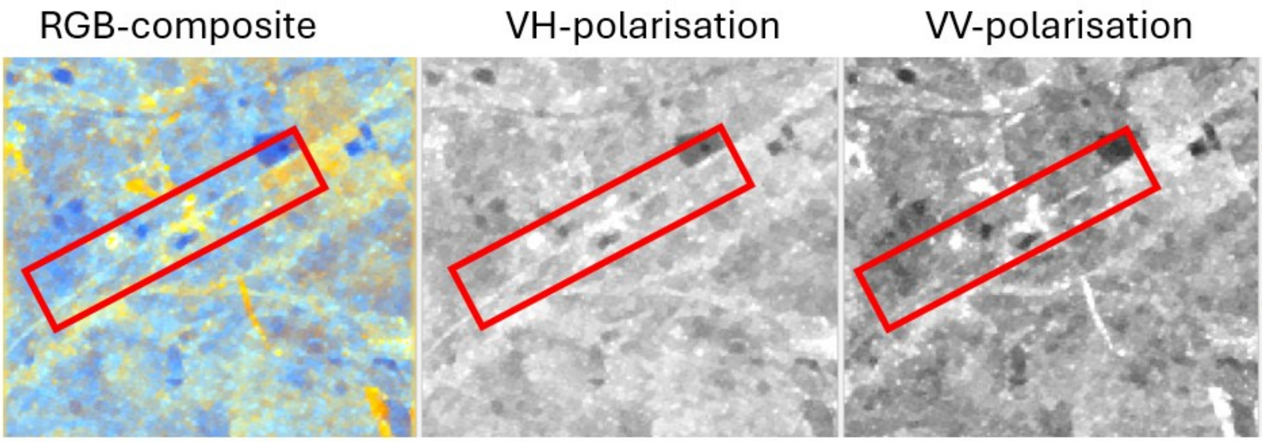


Figure 3.2 A railroad imaged across different polarizations.

In Figure 3.2, the VH-polarization exhibits the highest backscatter from the railroad this is due to the nature of the railroad containing complex geometries of stones and rail structures.

Although the dominant scattering mechanism for the VH-polarization is volume-scattering which is present in dense vegetation, the geometry and rocky nature of the rails cause double-bounce scattering which can depolarize the incoming vertical signal. As the VV-polarization is sensitive to surface scattering it does have a large amount of backscatter from the railroads but when compared to the VH-polarization it exhibits a lower backscatter, as the VH is more effective at receiving these geometries.

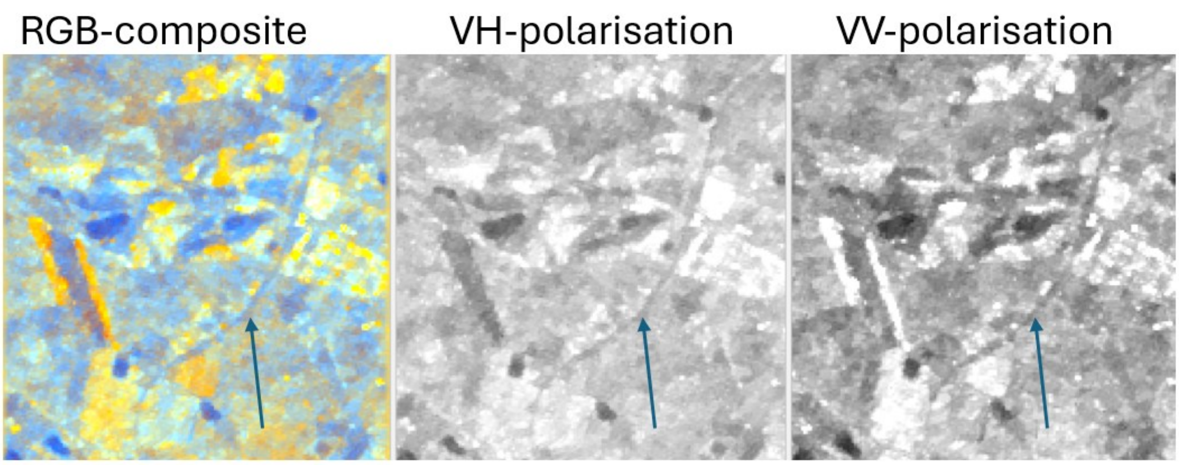


Figure 3.3 A road imaged using different polarizations.

In Figure 3.3, the VV-polarization exhibits very low backscatter this is a result of the VV-polarization. The primary scattering mechanism of the VV-polarization is surface scattering, as the road is generally smooth, there is very minimal backscatter over these regions. The VH-polarization does exhibit slightly more backscatter than the VV-polarization due to the this could be a result of roads containing gravel. If the gravel is the same as the wavelength of the C-band there could be some multipath effects that cause a subtle variation in the backscatter. The RGB composite image was able to detect the road in the blue channel this channel coincides with surface scattering as this is the ratio between the two polarizations.

3.3 Water

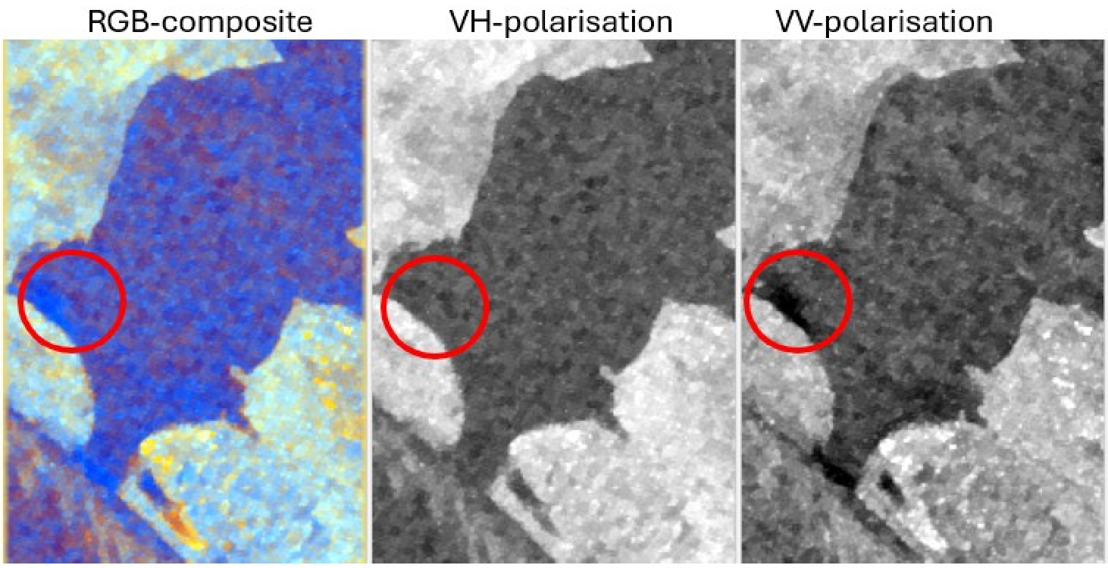


Figure 3.4 A waterbody imaged by different polarizations.

As water is a specular reflector it is expected that the waterbody should be completely black. Rather, in Figure 3.4 there are large amounts of backscatter in both polarizations. This could be that the waterbody is an estuary which experiences more turbulent water. As the water contains waves the incoming signal can have a multi-path off multiple waves and return to the sensor, this can act as a depolarizer which is why there is a higher backscatter in the VH-polarization. The area in the red circle depicts the difference in backscatter between the two polarizations, where the VV-polarization contains less speckle than the VH-polarization. The VV-polarization’s prominent scatterer is surface scattering which is why it can highlight areas with calmer waters. The RGB image detects the water in the blue channel which is where surface scattering is present.

3.4 Urban

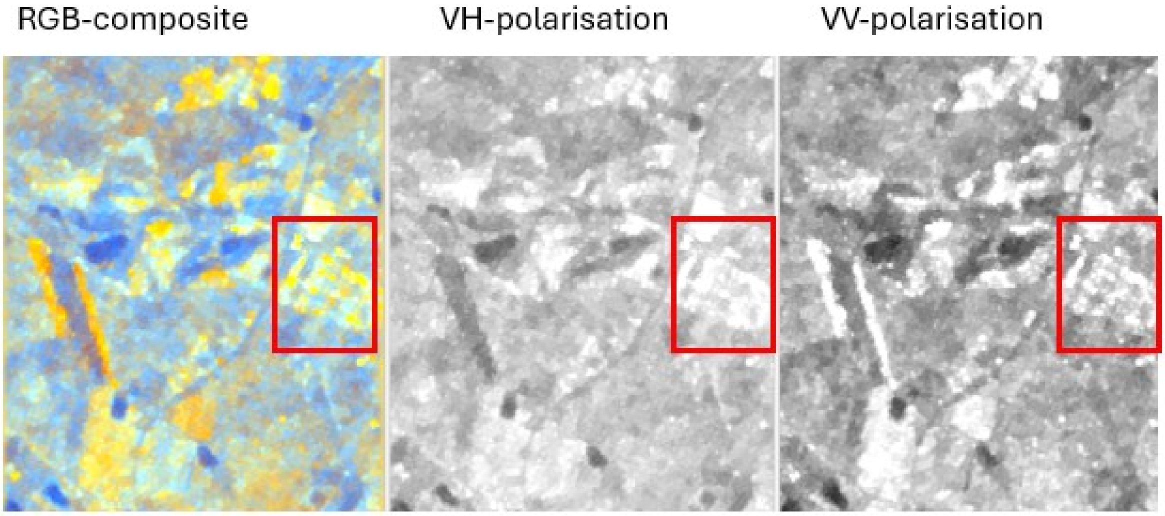


Figure 3.5 Urban area image by different polarizations.

As urban areas are made up of complex geometries, it allows for double-bounce scattering. The VV-polarization is expected to contain more backscatter than the HV-polarization as the vertical alignment of the buildings makes it prone to a double bounce from incoming vertical signals. Conversely, the HV polarization contains more backscatter than the VV polarization. This can be a result of depolarization of the V signal being received by the HV polarization. Depolarization is common in urban areas as there are multiple corners for the incoming signal to hit and be converted to the opposing polarization.

3.5 Terrain

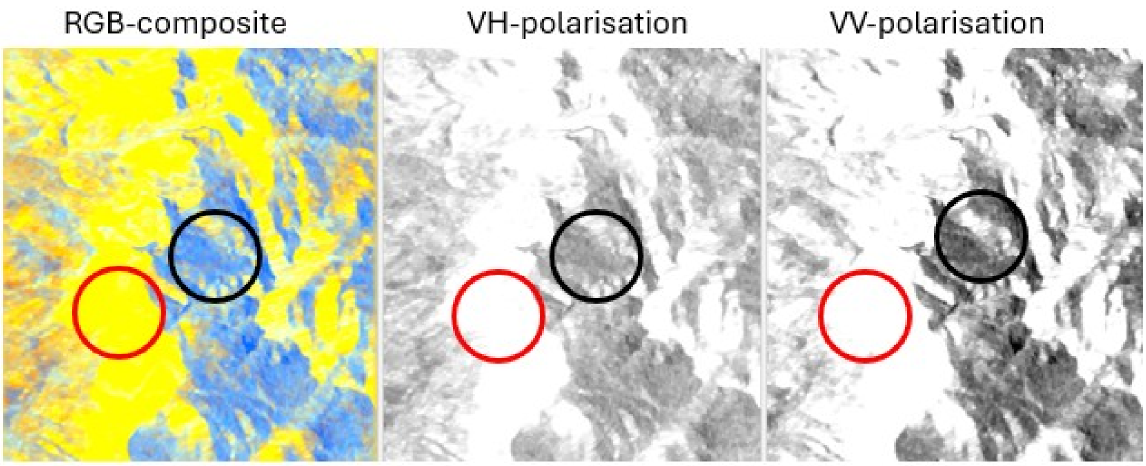


Figure 3.6 Steep terrain.

In Figure 3.5, the area within the red circle exhibits RADAR foreshortening as in both polarizations there are substantial amounts of backscatter. Conversely, in the black circle there is RADAR shadow which is depicted by low backscatter. This causes the terrain to tilt toward the sensor.

Task 4 (Comparison of C-band and X-band backscatter)

Thoroughly compare the two images and discuss the differences and similarities in the backscatter. Consider all the sensor parameters and ground parameters that might be different between the two images. These scenes will have vastly different spatial resolutions, which should be part of your discussion. You could also consider the time of day that the two scenes were captured, or any other differences between the two sensors / scenes. You may do this comparison in any software environment you prefer. Include annotated screenshots to highlight the differences you are discussing. [~450 words]

4.1 Spatial Resolution

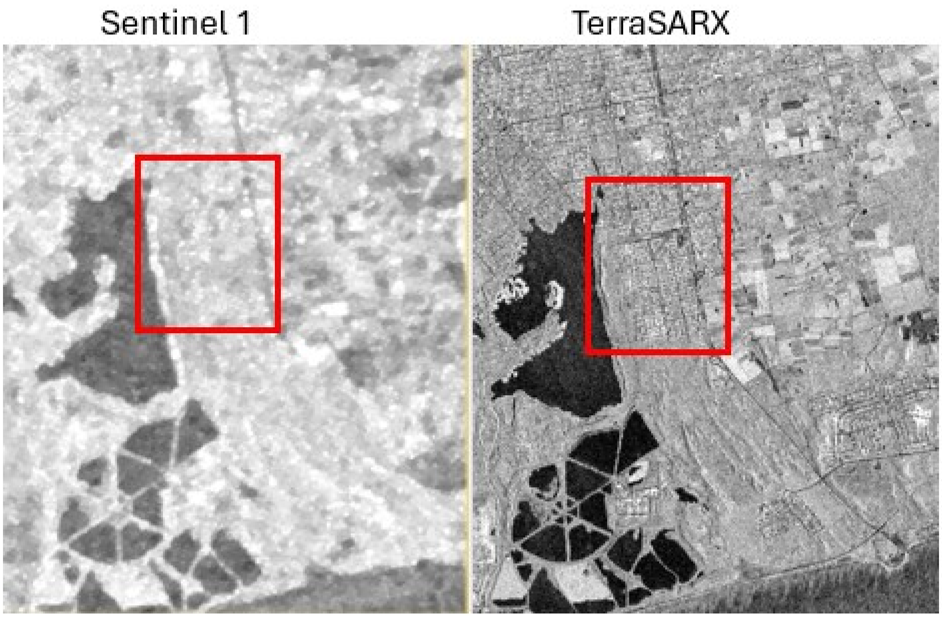


Figure 4.1 Image comparing the spatial Resolutions between the Sentinel 1 and TerraSARX sensors.

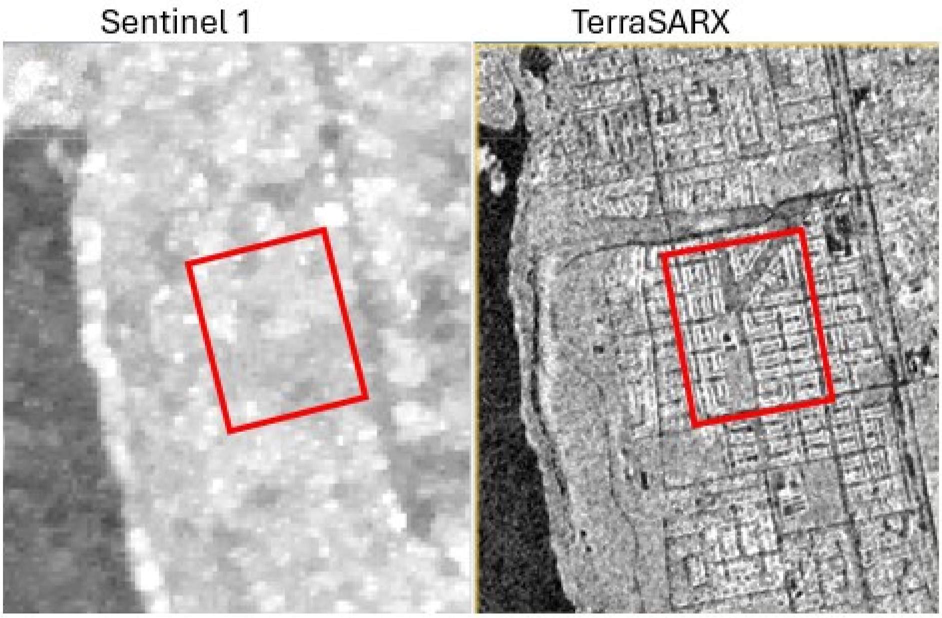


Figure 4.2 Image comparing the spatial Resolutions.

In Figure 4.1 and 4.2, the Terra SAR X image can capture a significantly larger amounts of detail when compared to the Sentinel 1 image. This is a Result of the TerraSAR X sensor capturing images at a spatial resolution of 3m, conversely the Sentinel 1 image has a spatial resolution of 20m. The lower spatial resolution creates mixed pixels as seen in Figure 4.2 where backscatter of multiple houses can be averaged into minimal pixels. The finer spatial resolution can preserve edges which create more detail in the image. This impacts the types of analyses done on the images, where TerraSAR X images could be used for fine detail classification whereas Sentinel 1 images would be used for general land cover mapping.

4.2 Wavelength

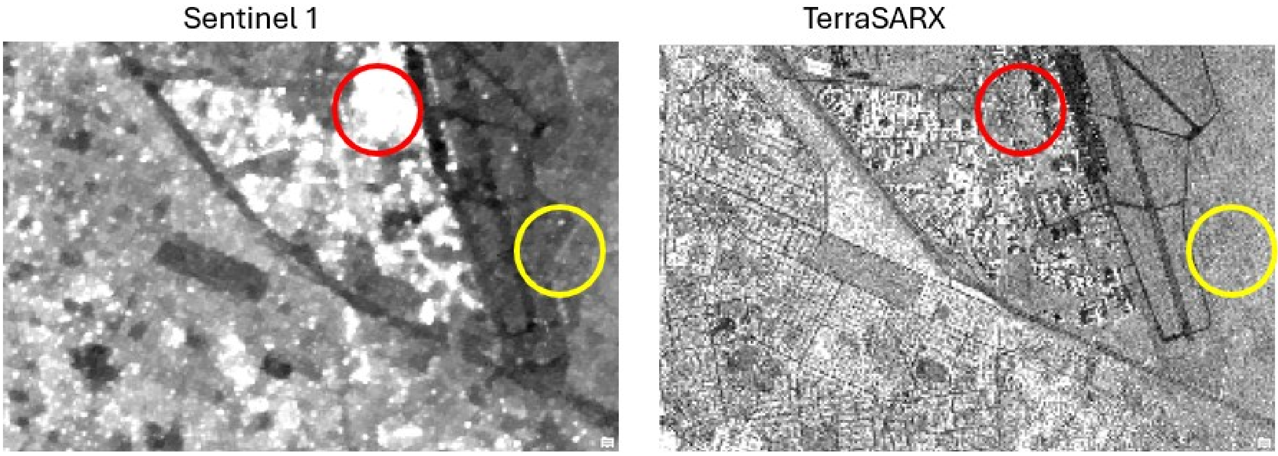


Figure 4.3 Shows an area with agricultural fields and an urban area compared between the two sensors.

As microwaves interact with objects similar to the size of the signal's wavelength. The TerraSAR X's X-band has a wavelength of 3cm, this results in the signal interacting with objects of that size which includes tree canopies and soil. The Sentinel 1's C-band has a wavelength of 5.6cm which interacts with objects such as tree canopies. Even though both sensors interact with the same objects the 3cm wavelength of the X-band can capture at a much finer spatial resolution than the C-band.

In Figure 4.3, urban area within the red circle has much larger backscatter intensity in the Sentinel 1 image than the TerraSAR X(TSX) image. This is due to the C-band being effective at double-bounce scattering off building, due to the spatial resolution of the sentinel 1 image, the backscatter is larger due to the averaging of the pixels.

The agricultural land within the yellow circle has larger backscatter in the TSX image than the Sentinel 1 image. This is due to the X-band only being able to capture the tops of the vegetation canopy which, as the wavelength is closer to the leaves, means the scattering is more intense. The C-band which can penetrate some leave does not return as significant backscatter.

4.3 Polarizations

The Sentinel 1 sensor is comprised of 2 sensor polarizations which are the VH and VV, whereas the TerraSARX is comprised of only one polarization being the HH.

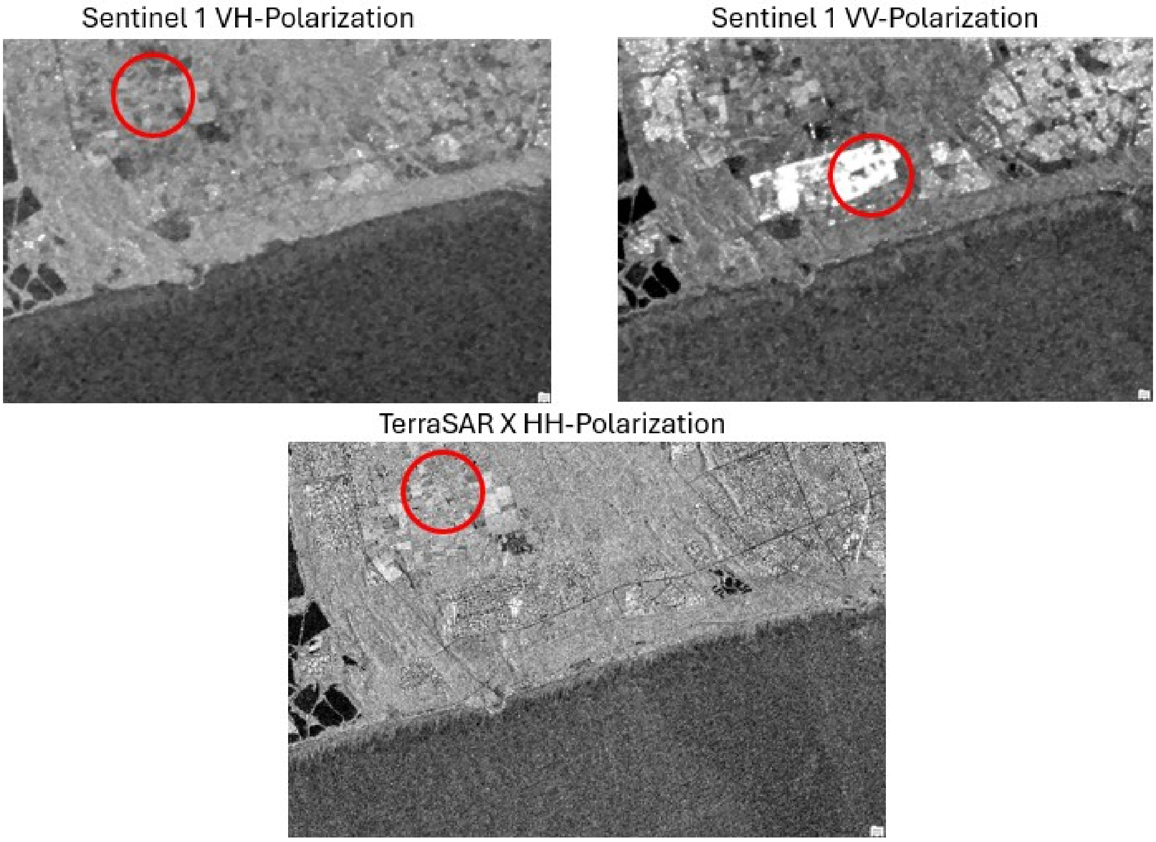


Figure 4.4 Polarization of a scene by 2 sensors.

The VH polarization is a cross-polarized signal, it is useful as it is highly sensitive to depolarization. The main scattering mechanism in this polarization is volume scattering, which is caused by vegetation canopies, as the signal interacts with multiple objects it causes depolarization. In Figure 4.4, the red circle highlights where the VH polarization was able to identify different densities of vegetation within the agricultural field.

The VV polarization is like polarized, its main scattering mechanism is surface scattering. Where smoother surfaces return minimal backscatter, the VV polarization receives substantial backscatter from vertical surfaces which are present in urban areas. In Figure 4.4 the red circle highlights the strong surface scattering from the buildings in that area

The HH-polarization is like polarized, the main scattering mechanism that it receives is double-bounce scattering off vertical structures and the ground. This polarization is useful for identifying man-made structures.

4.4 Look direction

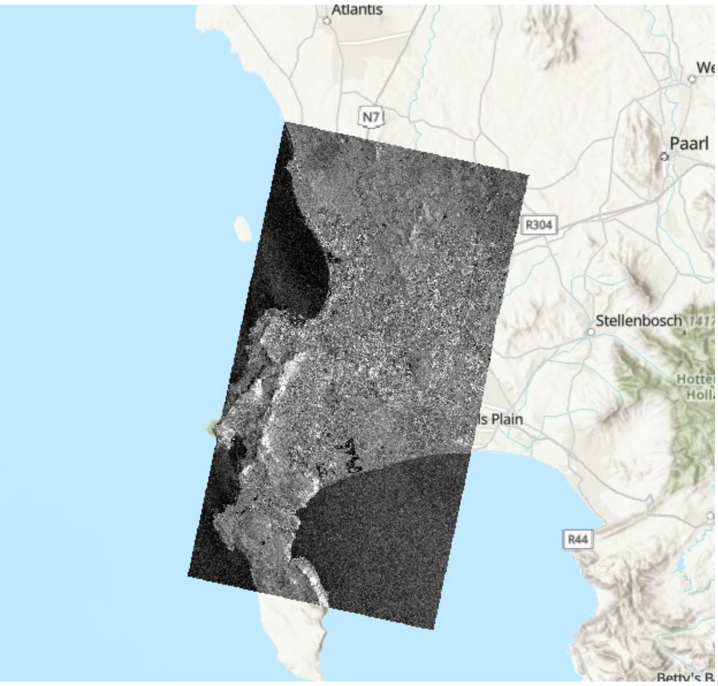


Figure 4.5 The whole scene of the TerraSAR X image.

In Figure 4.5, the image is slanted to the left, this indicates that the satellite pass type is descending. As the sensor is looking to the right of the flight path the look direction is west, this results in slopes or elevated terrain that face east will be foreshortened and slopes facing west will contain RADAR layover. The RADAR shadow will be in the west direction.

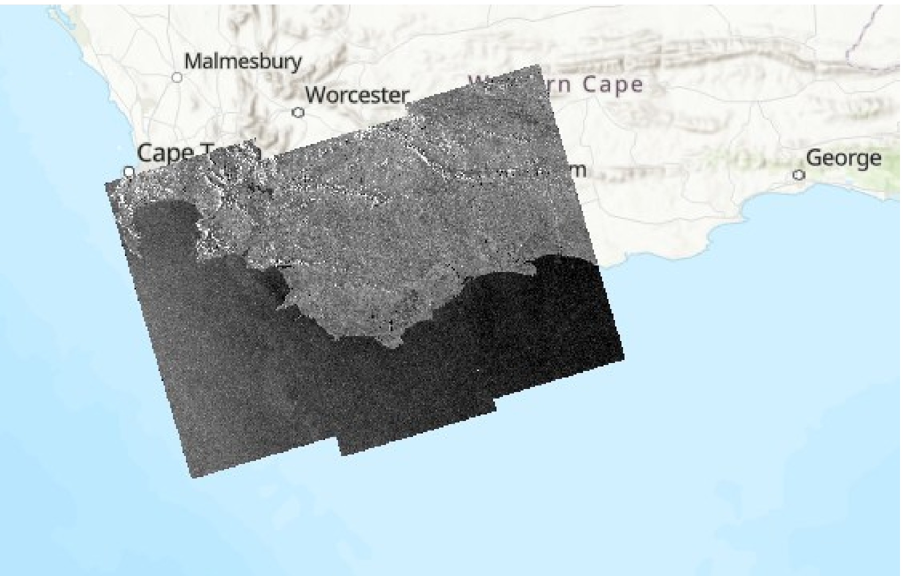


Figure 4.6 The whole scene of the Sentinel 1 image.

In Figure 4.6, the image is slanted to the right, this indicates that the satellite pass type is ascending. As the sensor is looking to the right of the flight path the look direction is east, this results in slopes or elevated terrain that face west will be foreshortened and slopes facing east will contain RADAR layover. The RADAR shadow will be in the east direction.

4.5 Acquisition times

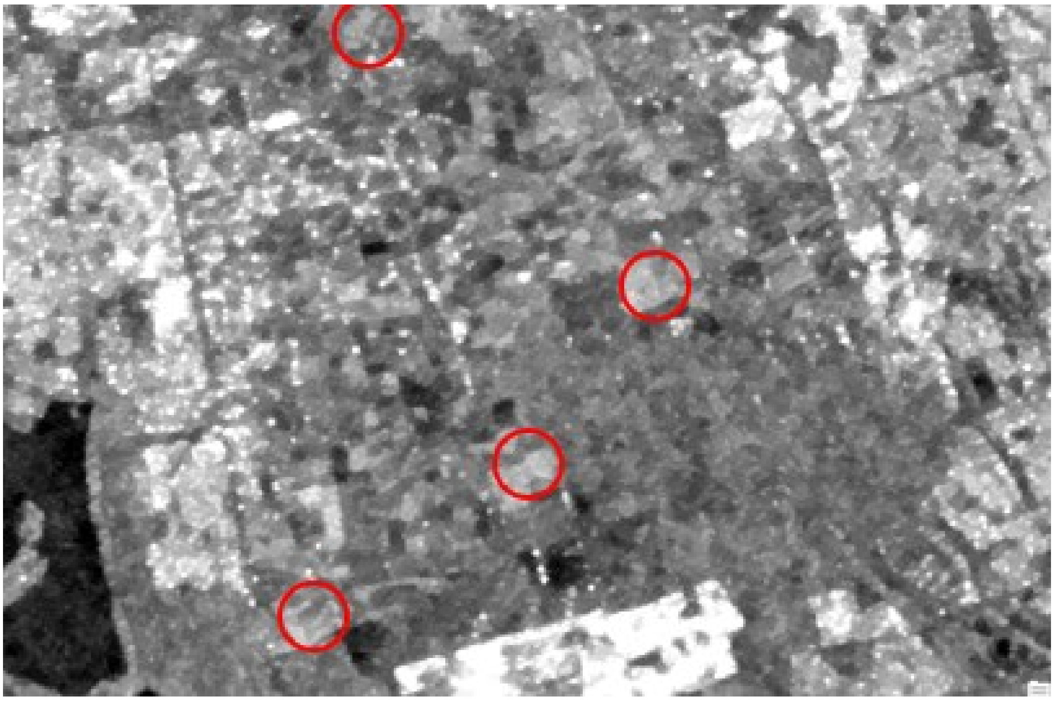


Figure 4.7 The VV-polarization of the Sentinel 1 image.

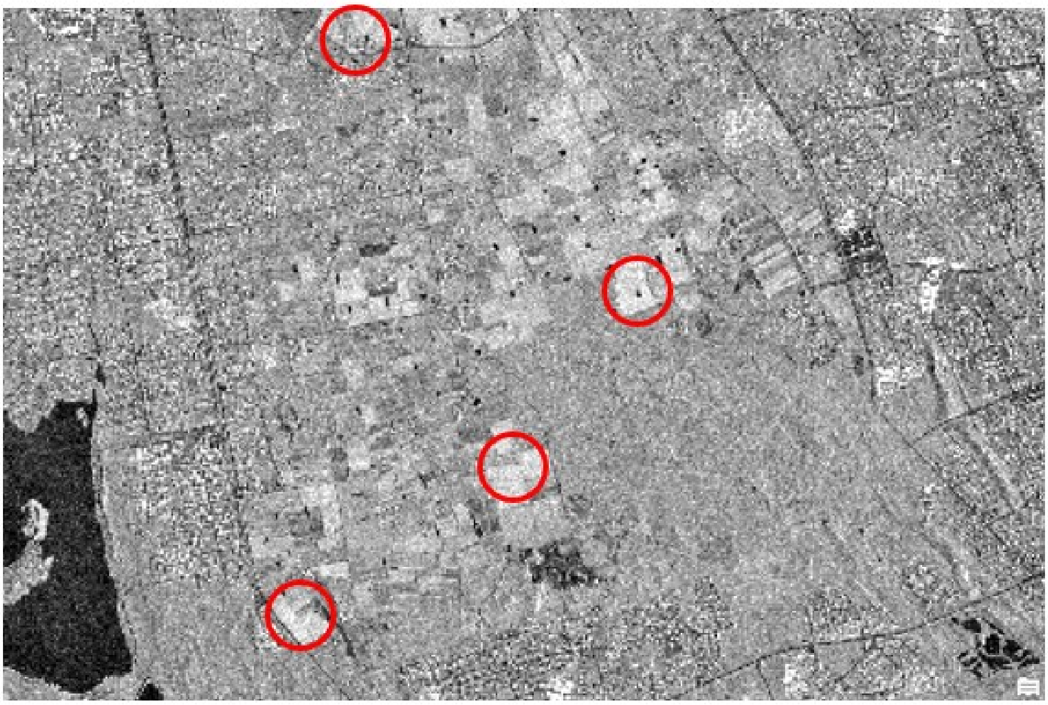


Figure 4.8 The HH-polarization of the TerraSARX image.

In Figure 4.8, the areas highlighted by the red circle contain high soil moisture due to the higher backscatter, in those same positions in Figure 4.7 those areas contain slightly less soil moisture. As the TerraSARX image was taken at 04:14 in the morning the soil moisture was substantially higher than the Sentinel 1 image that was taken at 18:34 in the evening.